Key Learning Points from VN2, the First Inventory Competition

Taking place from 29th September to 10th of November 2025, **VN2 is the first global inventory planning competition**.[1] Participants had to place six consecutive weekly orders for 599 Product x Store combinations, with the objective of minimizing holding and shortage costs. To do so, they received historical sales data (around 2-3 years per combination), information on historical shortages, and an anonymized master dataset.

Over 180 participants (mostly individuals, along with a few teams) submitted at least one order. They also shared around 20 notebooks with the community (including some end-to-end models).

In this article, I will share my key takeaways from hosting VN2 and present self-reported information shared by the top 20 participants. As for VN1, I will introduce a ranking of forecasting and inventory models and practices relying on my discussions with top competitors and my own experience.

Rank	Name	Total Cost (minus setup costs)	Cost reduction compared to benchmark	How much effort on forecasting vs. Inventory?	Forecasting Model	Inventory Model
1	Bartosz Szabłowski	3.763 €	13,2%	80%	Catboost	Week-3 forecast + safety margin - projected inventory. I translated the shortage and holding costs into a target service level of 83% (1/1+0.2), then into an equivalent z-score (0.97) using the standard normal distribution. The uncertainty term (sigma) is demand-dependent, increasing with the square root of the week-3 demand forecast and scaled by a factor phi. I optimized phi to 0.1 based on an initial validation to minimize total costs.
2	Matias Alvo	3.765 €	13,1%	30%	LGBM that predicts quantiles of cumulative demand for various horizons	Deep Reinforcement Learning method. Used MLP neural architecture that as input takes the state of inventory (units in stock and units arriving next week) and the predicted quantiles of cumulative future demand. (The method can also work without any explicit forecasts by providing relevant forecasting information to the neural policy directly). Built a simulator and optimized the weights of the neural network to minimize total cost on a train set.

[1] VN2 follows my first forecasting competition, VN1, which took place in 2024. The name, *VN*, stands for my initials.

3	Philip Stubbs, Jakub Figura	3.792 €	12,5%	70%	50% Seasonally-adjusted moving average, 50% LGBM	3-week forecast plus safety stock minus on hold and in-transit. Safety stock = k * RMSE, where k is optimized by simulation and is the same for all products. Reduction applied to the order if expected lost demand in the next two weeks. Simulation model calculating cost of forecast and buying strategy. We started with benchmark forecast model and buying to calculate cost of benchmark... and then made changes to see commercial difference
4	Carlo Cavalieri	3.794 €	12,5%	60%	DeepAR (RNN)	Project inventories using forecasted demand paths to select order quantities that minimize expected total costs, accounting for projected shortages before the order is received and holding costs in subsequent weeks. I optimize the model's hyperparameters to minimize costs in the pre-competition period.
5	Ruben van de Geer, Diederik Perdok, Simon Grest	3.824 €	11,8%	70%	LGBM model per quantile per horizon	Cost simulation based on probabilistic stock projection. We used forecasted quantiles to build a demand PMF for each store-product pair, then enumerated all possible future trajectories to compute the objective. Early on, we noticed consistent under-ordering, likely due to a bias in how the PMF was derived from the quantiles. To compensate, in rounds 7 and 8 we ordered using a higher percentile of the shortfall distribution, selecting the percentile that minimized expected costs over the remaining horizon.
26	Benchmark	4.334 €		95%	Moving Average with Seasonal Weights	4-week period coverage

Disclaimer. Most of my takeaways are based on self-reported information from the top 20 participants and my conversations with them. Despite my best efforts, some information may have been lost or misinterpreted along the way.

Evaluating Demand Forecasts vs. Inventory Policies

Over the last years, the demand forecasters enjoyed multiple forecasting competitions: Corporación Favorita in 2017, M5 in 2020, Intermarché in 2021, and, finally, VN1 in 2024. Each forecasting competition brought more learning points about what works and what doesn't, and kept up with technological developments. It's easy to assess the quality of a forecast – even if we could have some debates about the exact metric to use. **Evaluating the quality of inventory policies or decisions is much more complicated**. An inventory policy is *a set of rules for determining inventory decisions.*

I like to illustrate this to my students with the following example: Imagine we stand in a coffee shop. I forecast that they will sell 35 cappuccinos; You forecast that they will sell 25. At the end of the day, they made 29 cappuccinos. You made the best forecast. It's quite straightforward.

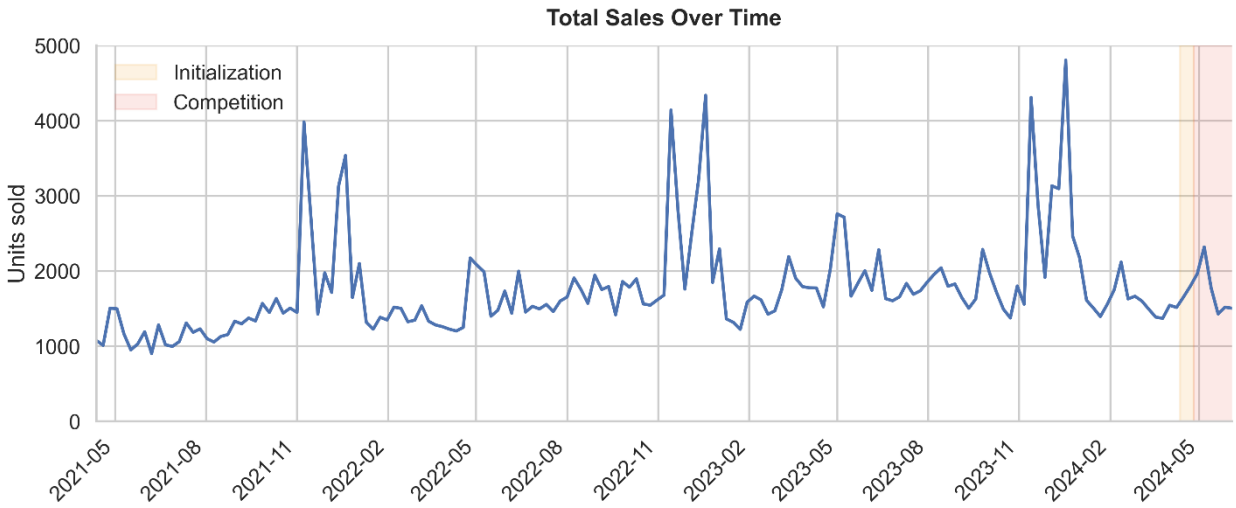
Now, let's imagine the same example with inventory decisions. I think they should order enough coffee beans to make around 1000 cups, but you think they should order only 500 cups. At the end of the day, they brew 150 cups. Which decision was best? Maybe I got a discount from the supplier by ordering more, and I reduced overall handling time by making a single big order. Maybe you did account for the upcoming holiday (the store will be exceptionally closed), or account for the fact that the supplier reliably delivers orders within 48 hours. Maybe we both overlooked the fact that the demand forecasts are quite reliable, etc.

The only way to assess which inventory policy is best would be to observe, over time, which one results in the lowest total cost.[2]

That's how I designed VN2: a competition over 6 rounds in which competitors had to place weekly orders to minimize total costs while facing the consequences of ordering too much or too little.

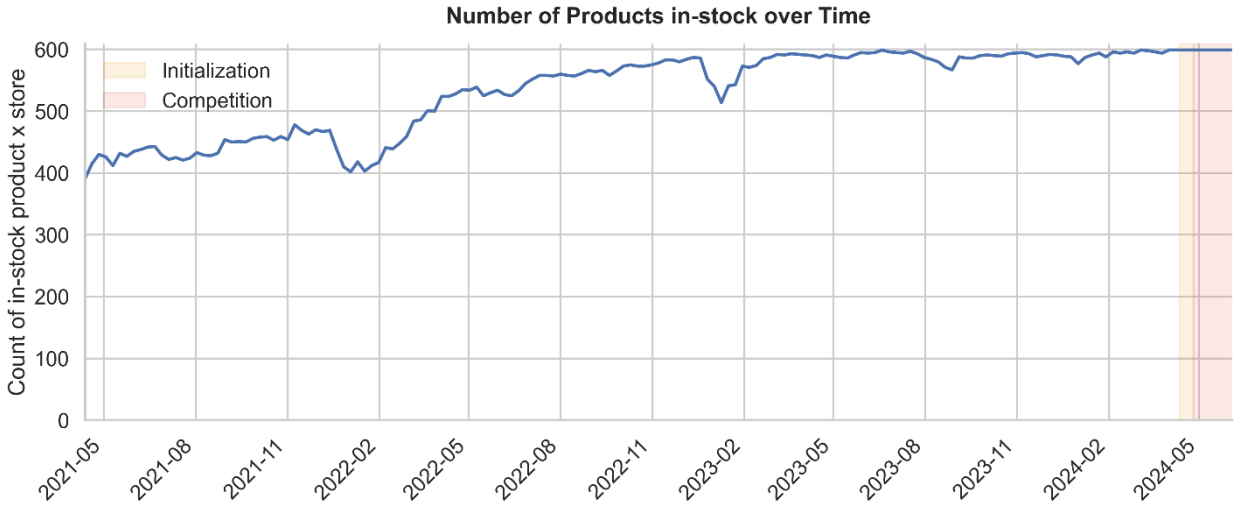
Competition Structure and Data

Before discussing the results and the best models, let's recap the underlying dataset: VN2 includes 297 Products and 67 Stores, for a total of 599 combinations. The data comes from an anonymous retailer. Participants began on Monday, April 15th, 2024, with access to historical sales and had to place their first orders. Since it takes two weeks to receive an order, we had an initialization phase (orange) covering the two weeks starting on the 15th of April. In other words, because it takes 2 weeks to get the order resulting from their first decision, participants had no impact on the inventory levels during the initialization phase. So, they were measured over the consecutive 6 weeks (red).

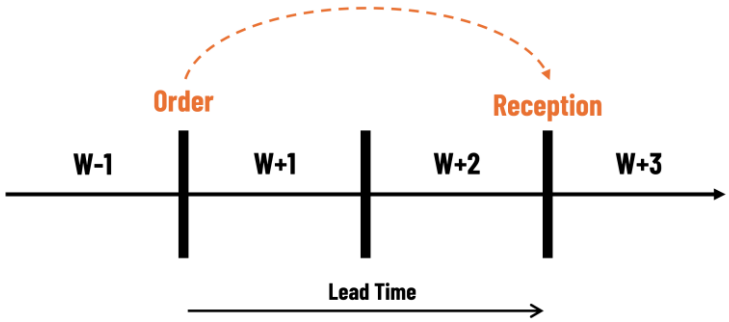


Most products suffered historical shortages, so participants received historical indicators highlighting *likely* shortages. These are *likely* shortages, as the anonymous company sharing VN2 data reported no inventory at the end of a given week – but inventory figures could be wrong and the fact that inventory dropped to zero doesn't mean for sure that they suffered from lost sales. As shown in the following graph, many products experienced shortages during or immediately after the season peaks.

[2] We assume here that we want to minimize total costs (including the opportunity cost of losing sales – so that *minimizing total costs* is equivalent to *maximizing profits*). Unfortunately, in practice, most supply chains can't map out relevant costs, so their objective might sound like "Deliver on service level targets using the lowest inventory level possible."



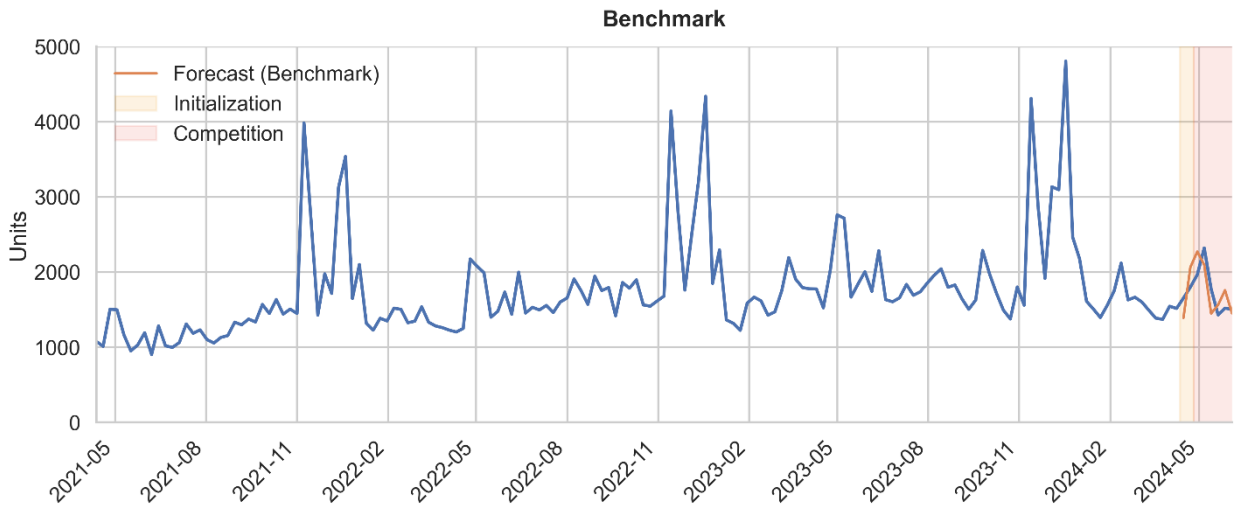
Finally, the VN2 supply chain operated on weekly orders with a 2-week lead time.



Can you Beat the Benchmark?

As was the case in VN1, a simple benchmark was also proposed in VN2 for participants to beat. The simple benchmark was a 13-week moving average with seasonal components (computed at the global levels) combined with a 4-week stock target. It takes around 50 lines of Python code to run this simple benchmark (excluding graphs). You can download the codebase to reproduce the benchmark here.

This benchmark wasn't optimized or tested in any way – the period coverage is an arbitrary number.



The top 5 competitors beat the benchmark by an average of 13%. **But only 25 participants (out of 180!) could beat it.** Most end-to-end forecasting/inventory models failed to deliver any value. This is **the first key learning from this competition: it's exceptionally difficult for most practitioners to beat a simple yet smart benchmark**. I frequently observe this with my clients: those who don't compare their forecasting accuracy against a moving-average benchmark often find they don't beat it.

Based on the feedback from competitors (and by further analyzing the underlying forecasts they did), it seems that (by chance!) the seasonal moving average used in the benchmark was an accurate model. On the other side, the 4-week period-coverage wasn't optimal and could be tuned to reduce costs. If you were to follow the method I proposed in my Udemy course, *Inventory Planning - How to Optimize Inventory Policies*, to optimize the period-coverage, you would get a period-coverage of around 3.5 weeks, resulting in a total cost of around 3,900€. This method would have ranked top 6.

Rank	Name	Total Cost in Euro (minus setup costs)	Cost reduction compared to benchmark	How much effort on forecasting vs. inventory?	How many lines of code are required to generate an order?
1	Bartosz Szabłowski	3,763 €	13.2%	80%	600
2	Matias Alvo	3,765 €	13.1%	30%	100
3	Philip Stubbs, Jakub Figura	3,792 €	12.5%	70%	150
4	Carlo Cavalieri	3,794 €	12.5%	60%	250
5	Ruben van de Geer, Diederik Perdok, Simon Grest	3,824 €	11.8%	70%	2,000
6	Mohammad Abdollahi	3,955 €	8.7%	80%	2,000
7	Richard Plat	3,972 €	8.3%		1,500
8	Pavel Novikov	3,981 €	8.1%	100%	100
9	Antoine Schwartz	3,989 €	8.0%	90%	200
10	An Hoang	4,039 €	6.8%		10
11	Victor Voltz	4,071 €	6.1%		
12	Olivier Jonard	4,109 €	5.2%	10%	
13	Johann Robette	4,119 €	5.0%		4,000
14	Adem Akgül	4,125 €	4.8%	80%	60
15	Shuyang Zhang	4,146 €	4.3%	20%	200
16	Philipp Hallmeier	4,159 €	4.0%	20%	1,000
17	Arsa Nikzad	4,174 €	3.7%	50%	150
18	John Woodward	4,175 €	3.7%	5%	1,500
19	Andrew Henrey	4,207 €	2.9%	5%	650
20	Motoki Saitama	4,238 €	2.2%	20%	20
26	Benchmark	4,334 €		95%	55

Forecasting Models

Let's evaluate the models by ranking them from A-tier (*best*) to D-tier (*worst*). A-tier models (or practices) are the ones I consider the most likely to consistently deliver value - without too much work. D-tier models and practices, on the contrary, are nearly guaranteed to fail.

First, let's have a look at the top models for VN2.

Disclaimer. The fact that a model was successful or not during VN2 is not a definitive demonstration of its effectiveness or its lack of it. Results may be influenced by survival bias: many participants might have tried a specific

technique, and, by chance, a few of them made it to the top. Alternatively, it is possible that a new model or technique may have been implemented by only a few participants, who failed to integrate it into an end-to-end pipeline, so we don't find it among the top.

Rank	Name	Forecasting Model	Forecast Type	Lines of code to create an order?	Lines of code to optimize the model?	Minutes to create an order?	Minutes to optimize?
1	Bartosz Szabłowski	Catboost	Mean	600	800	240	210
2	Matias Alvo	LGBM predicting quantiles of cumulative demand for various horizons	Quantile	100	1.000	-	12
3	Philip Stubbs, Jakub Figura	50% Seasonally-adjusted moving average, 50% LGBM	Mean, asymmetric loss penalization for LGBM	150	283	1	2
4	Carlo Cavalieri	DeepAR (RNN)	Distribution	250		2	
5	Ruben van de Geer, Diederik Perdok, Simon Grest	LGBM model per quantile per horizon	Distribution (based on various quantiles)	2,000	500	5	1
6	Mohammad Abdollahi	A combination and ensembling of stat (Exponential Smoothing, ARIMA, Naives), ML (LGBM), and DL (N-BEATs/N-HiTS, DeepAR) models as local and global models to compete.	Mean	2,000		2	350
7	Richard Plat	Selection of the best Stat, ML, and DL forecasting models per SKU, then ensembling of the best Stat, ML, and DL. Most common models: AutoDLinear, MA, LGBM, NHITS, NBEATS, CatBoost, ARIMA.	Mean	1,500	1.500	720	720
8	Pavel Novikov	Exponential Smoothing	Mean	100	100	0	0
9	Antoine Schwartz	Chronos 2	Quantile	200	400	5	120
10	An Hoang	LGBM	Mean + Conformal Predictions	10	2.000	10	120
11	Victor Voltz	LGBM	Quantile				
12	Olivier Jonard	8 weeks moving average	Mean			0	
13	Johann Robette	Empirical distribution, Probabilistic forecast	Distribution	4,000		1	
14	Adem Akgül	Catboost	Mean	60	60	40	240
15	Shuyang Zhang	Empirical distribution	Quantile	200	50	0	
16	Philipp Hallmeier	LGBM	Quantile	1,000		5	
17	Arsa Nikzad	Ensembling of stat models	Mean	150		15	
18	John Woodward	Holt-Winters (Additive), exponentially weighted MA	Mean and Quantiles based on normal-banded forecast error	1,500	150	0	5
19	Andrew Henrey	LGBM with a horrible vibe-coded bug that made my forecasts quite poor	Distribution	650		120	
20	Motoki Saitama	Arima	Mean	20	20	60	60
26	Benchmark	Moving Average with Seasonal Weights	Mean	55	800	1	

A-Tier

- **Ensembling models**. Ensembling models means averaging various forecasts (usually from different models or using different random seeds) into a single forecast. As you combine different forecasts into a single one, you get an accuracy boost.[3] The trick is to properly assess the weights assigned to each model (allocating flat weights to all models is unlikely to deliver much value). Ensembling models comes at the cost of increased complexity (you have to build and optimize multiple models) and higher running costs. It works great for winning competitions – top competitors love to experiment with different models! But it is

[3] This is similar to using the wisdom of the crowd for (forecasting) models!

not frequently used in practice by software vendors. At SupChains, we've been relying on ensembling since 2018 (at the time with neural networks) to boost our models' accuracy while stabilizing our forecasts.

- **LGBM**. LGBM is a machine learning model that is frequently used for forecasting tasks. It's fast, reliable, and (relatively) simple to use. Moreover, it can take many different variables and business drivers as inputs. It's SupChains' favorite, was VN1 top competitors' favorite, and is still among VN2's top solutions. That said, the key success factor with LGBM – as with most, if not all, ML models – is feature engineering (which data you feed *the model and how*) and hyperparameter optimization.
- **Accounting for Shortages.** Accounting for shortages is key to supply chain planning and forecasting models. VN2 was filled with holes in sales data due to shortages. Top participants found ways around this. At SupChains we also spend time with our clients to process their inventory data to spot shortages.

B-Tier

- **CatBoost**. Very similar to LGBM, CatBoost has been successfully used by several top competitors. I personally do not use it because, in my (limited) experience, it doesn't offer added value over LGBM and lacks its reliability.
- **Transformers**. As for VN1, one of the top participants successfully used a Transformer forecasting model as a forecast engine. If you accept using a pre-trained, black-box, closed model, transformers can deliver quite good accuracy.

C-Tier

- **Seasonal Moving Averages.** As demonstrated by the VN2 benchmark, a good moving average (accounting for shortages) combined with seasonal weights can go a long way.
- **Statistical Models.** If properly selected, optimized, and within an ensemble, statistical models can deliver value. Unfortunately, it takes great skills to be able to deliver on these requirements, and statistical models will struggle with business drivers such as promotions or pricing (and shortages).

D-Tier

- **Outlier Detection**. Only two single top participants reported flagging and excluding outliers. At SupChains, we do not detect or clean outliers. Instead, we invest time cleaning input data with our clients and then include business drivers as inputs to our model. After these steps, there are usually no outliers remaining.
- **ARIMA**. Compared to most statistical models, ARIMA is slow, difficult to optimize, and doesn't provide better forecasts or extra features compared to any other statistical model.

A	**Ensembling**	**LGBM**	**Accounting for Shortages**
B	**Catboost**	**Transformers**	

C	Statistical models	Seasonal Moving Averages
D	Outlier detection	ARIMA

Inventory Models

Now that we've reviewed the forecasting models, let's examine the inventory models used by VN2 participants and provide an overall ranking.

Top participants used various techniques to deliver great value: mean forecasts (also referred to as *point-forecasts)* used with period coverages, quantile ordering, distribution forecasts (also referred to as *probabilistic forecasts*) with inventory projection, advanced inventory optimization engines, and so on. It seems that many different techniques can deliver amazing inventory policies.

Despite this great variety in winning solutions, **some practices never made it to the top** (DDMRP, ABC, and *Intermittent, Lumpy, Erratic, Smooth* segmentations), and no participant using them could beat the simple benchmark. They can be safely ranked as worst practices.

When discussing inventory models, I will focus on the underlying sub-components (framework, concepts, ideas) that can then be combined into different models.

If you want more information about the underlying models or concepts, see my book, *Inventory Optimization: Models and Simulations*.

Rank	Name	Total Cost in Euro (minus setup costs)	How much effort on forecasting vs. Inventory?	Inventory Model
1	Bartosz Szabłowski	3,763 €	80%	Week-3 forecast + safety margin - projected inventory. I translated the shortage and holding costs into a target service level of 83% (1/1+0.2), then into an equivalent z-score (0.97) using the standard normal distribution. The uncertainty term (sigma) is demand-dependent, increasing with the square root of the week-3 demand forecast and scaled by a factor phi. I optimized phi to 0.1 based on an initial validation to minimize total costs.
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3	Philip Stubbs, Jakub Figura	3,792 €	70%	3-week forecast plus safety stock minus on hold and in transit. Safety stock = k * RMSE, where k is optimized by simulation and is the same for all products. Reduction applied to the order if expected lost demand in the next two weeks. Simulation model calculating cost of forecast and buying strategy. We started with benchmark forecast model and buying to calculate cost of benchmark... and then made changes to see commercial difference
4	Carlo Cavalieri	3,794 €	60%	Project inventories using forecasted demand paths to select order quantities that minimize expected total costs, accounting for projected shortages before the order is received and holding costs in subsequent

				weeks. I optimize the model's hyperparameters to minimize costs in the pre-competition period.
5	Ruben van de Geer, Diederik Perdok, Simon Grest	3,824 €	70%	Cost simulation based on probabilistic stock projection. We used forecasted quantiles to build a demand PMF for each store–product pair, then enumerated all possible future trajectories to compute the objective. Early on, we noticed consistent under-ordering, likely due to a bias in how the PMF was derived from the quantiles. To compensate, in rounds 7 and 8 we ordered using a higher percentile of the shortfall distribution, selecting the percentile that minimized expected costs over the remaining horizon.
6	Mohammad Abdollahi	3,955 €	80%	3-week Forecast minus net inventory. No safety stocks. Order capped based on recent sales (median & p95).
7	Richard Plat	3,972 €		3-week Forecast minus net inventory. No safety stocks.
8	Pavel Novikov	3,981 €	100%	Matched 3-week inventory to 3-week forecast and set minimum 3-week inventory to 1
9	Antoine Schwartz	3,989 €	95%	69% quantile ordering
10	An Hoang	4,039 €		LGBM /EnBPI to create probabilistic forecast + Simulation to find the optimal order quantity.
11	Victor Voltz	4,071 €		Quantile ordering
12	Olivier Jonard	4,109 €		Safety stock based on 83% quantile (extrapolated from the last 17 weeks)
13	Johann Robette	4,119 €	10%	Cost simulation based on probabilistic stock projection
14	Adem Akgül	4,125 €	80%	Selection of one of 3 policies (Static, Hybrid, or Dynamic)
15	Shuyang Zhang	4,146 €	20%	Empirical demand distribution / stochastic programming
16	Philipp Hallmeier	4,159 €	20%	83% quantile ordering
17	Arsa Nikzad	4,174 €	50%	83% quantile ordering via bootstrap on out-of-sample error per horizon
18	John Woodward	4,175 €	5%	Forecast-based Percentile-Guided Inventory Optimization with ABC tuning
19	Andrew Henrey	4,207 €	5%	Dynamic Programming
20	Motoki Saitama	4,238 €	20%	Direct optimization based on the forecast
26	Benchmark	4,334 €	95%	Period coverage

A-Tier

- **Optimization Framework**. The one key factor of winning solutions is that competitors had a way to evaluate their policies (using historical data) and tune them.
- **Distribution and Mean Forecasts**. Both probabilistic and mean forecasts could deliver great results in VN2.
- **Inventory Projection Ordering**. Being able to project how much inventory you'll have in a few weeks is key to knowing how much to order today. Especially with seasonal demand and potential shortages along the way. Top participants either used point-forecasts or forecast distributions to project their inventory forward. The ability to project inventory is also key to estimate the long-term impact of overordering – especially if combined with probabilistic forecasts.

Table 1 Example of 3-week ahead inventory projection with short-term shortages.

Period	Week 1	Week 2	Week 3	Week 4	Week 5
Starting Inventory	100	0	0		
Receipts	20	25	?		
Forecast	150	100	50	80	50
Expected Shortages	30	75			
Ending Inventory	0	0			

B-Tier

- **Net Inventory Level Ordering.** Inventory net level is computed as the current on-hand inventory plus incoming orders. You can place orders by comparing this net level against your reorder points or up-to-

levels (following an R,S, or R,s,Q policy). Unfortunately, net inventory-level ordering doesn't account for short-term expected shortages, resulting in over-ordering.

In Table 1, using an inventory projection, to achieve a safety stock of 1 week of forecast coverage at the end of W3, you need to order 130 pieces. Using a 4-week coverage (R,S) inventory policy based on the net inventory level (without shortages, this policy would also result in 1 week of safety stock), you would order 235 units (150+150+50+80 units of forecast – (100+20+25) of net inventory units).

- **Reinforcement Learning (RL)**. This amazing technique was only used by Matias Alvo who reached the second position. Using RL, you train a neural network to directly make inventory decisions based on the current state (inventory level, incoming shipments) and historical demand patterns. Despite this solution achieving an amazing score, its complexity and training time keep it niche. We'll see if this technique gains wider adoption in the coming years, but its current complexity is likely to be a roadblock.

C-Tier

- **Quantile Ordering (Newsvendor)**. Many participants forecasted a single probability-based quantity (that I also refer to as *quantile forecasting*). For instance, forecasting 100 units at the 80% quantile means that demand is expected to be below 100 units in 80% of cases. Forecasting quantiles is straightforward with most ML models, which can directly predict any quantile. You can then use this forecast to make your order. Participants tried out many variations of this quantile ordering (some better than others): some computed the 3-week cumulative quantile, some used the theoretical newsvendor quantile (which isn't adequate for the VN2 setup, as the newsvendor model assumes that you can get rid of the leftovers), some optimized the quantile using historical simulations, etc.

  I ranked quantile ordering as C-tier, as it didn't reach the top 5 of the competition despite the fact that many participants tried it. And it suffers from two main flaws. First, using quantile ordering at the start of a high season will likely result with piles of leftovers that could be stuck in your inventory for quite some time. Second, quantile forecasts can't be used to create an inventory projection. Some participants used an optimization framework to determine the best quantile, whereas others stuck to the usual newsvendor optimum formula.
- **Safety Stock Formula (using the forecasting error)**. Some participants defined safety stocks as a constant time historical forecast error. And optimized the constant using historical simulations.

$$Safety\ Stocks = k \times RMSE\ (or\ MAE)$$

  Somehow, this is similar to defining

$$Safety\ Stocks = z \times RMSE \times \sqrt{L + R}$$

  But simpler as you simply optimize *k* and do not refer to *z* and its underlying cycle service level.

D-Tier

- **ABC Segmentation**. ABC Segmentation (setting inventory targets based on historical demand volumes) is a technique in direct opposition to using an optimization framework (that I ranked as A-tier) and that doesn't account for lead time variability, profitability, forecast errors, or expected future demand (high or low). I don't see any reason why such a poor technique, with no foundation, should be used in practice.
- **Intermittent, Lumpy, Erratic, Smooth**. Same as ABC Segmentation.
- **DDMRP**. DDMRP fails to forecast demand (especially if there are trends, seasonality, shortages, or promotions – that is, any usual supply chain demand pattern), and doesn't provide any framework to optimize its parameters.

- **Safety Stock Formula (with demand variation)**. Finally, some participants used the usual safety stock formula:

$$Safety\ Stocks = z\sigma_d\sqrt{L+R}$$

This formula is worse than the C-tier one as it doesn't account for forecast errors, but only for demand variability (you can face high demand variability and enjoy low forecast errors). Moreover, z is tight to the *cycle service level*, a metric that doesn't correlate well to other service level metrics such as fill rate (see [Inventory Optimization: Models and Simulations]).

A	**Optimization Framework**	**Inventory Projection**	**Mean Forecasts**	**Distribution Forecasts**
B	**Net Inventory Level Ordering**	**Reinforcement Learning**		
C	$k \times RMSE$	**Quantile Ordering**		
D	**DDMRP**	**ABC Segmentation**	**Intermittent, Lumpy, Erratic, Smooth**	$z\sigma_d\sqrt{L+R}$

Key Lessons Learned

As we saw for both forecasting and inventory models, the one key skill of top competitors is that they can evaluate and optimize their models at scale. For forecasting, it usually means using cross-validation models using temporal sliding windows. For inventory, it usually means evaluating models (and tweaking parameters) using historical simulations.

Regarding models, we find the same top forecasting models (*ML in general and LGBM specifically*) and best practices (*ensembling*) as for VN1. Whereas, for inventory models, we see that different approaches can result in great results (*point or distribution forecasts, reinforcement-learning, random simulations, etc.*) – even if some approaches couldn't deliver value (*ABC Segmentation, demand-variability-based safety stocks, DDMRP*).

Overall, if you combine a well-designed, robust evaluation framework that you can use at scale (for both forecasting and inventory models) with various sub-models to test (including machine learning), you should be heading towards great inventory decisions.

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